# TNT Paper Kahoot: answer key

Paper 1: Sadiya, Alhanai, Ghassemi (2021), artifact review. Paper 2: Ravishankar et al. (2026), handwriting decoding.

**1. Paper 1 (Sadiya et al. 2021) is mainly a...**
**Review of artifact detection & correction methods (correct)**
New cheap EEG headset design
Motor imagery dataset
Handwriting decoding study
*Why:* It surveys how EEG artifacts are found and removed; it reports no decoding accuracies.

**2. Which of these is NOT an artifact-removal method covered in Paper 1?**
ICA (independent component analysis)
Empirical mode decomposition
Blind source separation
**Common spatial patterns (CSP) (correct)**
*Why:* CSP is a feature method for motor imagery, not an artifact cleaner.

**3. Paper 1 measures how well artifacts are REMOVED with...**
Classification accuracy
**RMSE / MSE (correct)**
Number of electrodes
Sampling rate
*Why:* Detection uses F1; correction quality uses RMSE, MSE and normalized MSE.

**4. In Paper 2, participants wrote or imagined which 4 letters?**
A, B, C, D
**L, O, V, W (correct)**
X, Y, Z, Q
E, F, G, H
*Why:* The dataset is called LOVW-8.5K.

**5. Chance level for Paper 2's 4-letter task?**
50%
10%
**25% (correct)**
3.8%
*Why:* 1 out of 4 = 25%.

**6. A study claimed 94%. Using ONE eye-movement component, Paper 2 got about...**
25%
50%
10%
**85.6% (correct)**
*Why:* Most of that 94% came from eye movements, not the motor cortex.

**7. Why did Paper 2 use time-ordered (unshuffled) folds?**
**So the model can't learn recording time (correct)**
To train faster
To get more data
To remove blinks
*Why:* Shuffled folds let a model learn slow drift over the session instead of the letter.

**8. Which model did BEST in Paper 2?**

REVE (big pretrained model)
**EEGNet (small specialist) (correct)**
CBraMod (big pretrained model)
MIRepNet
*Why:* EEGNet got 41.3%; the big foundation models did worse.

**9. Imagined writing, tested on a NEW session, scored...**
About 78%
About 41%
**Not above chance (correct)**
About 95%
*Why:* No cross-session transfer, even on 32-channel lab EEG.

**10. Paper 2's key claim?**
Bigger models always win
Cheap EEG beats lab EEG
Foundation models solve EEG
**Preprocessing can't fix a confounded design (correct)**
*Why:* If the design is confounded, the classifier still finds the leftover artifacts.

**11. How do the two papers fit together for our project?**
**Paper 1 = toolbox, Paper 2 = warning (correct)**
Both are headset reviews
Both report MI ceilings
They disagree about ICA
*Why:* Design out confounds first, then clean, then prove the signal is neural.